

ITEC 630 – List of Software Development Project Ideas – (not exhaustive, indicative only)

Web Development Projects

Project Title	Technologies Used
Personal Portfolio Website	HTML, CSS, JavaScript
E-commerce Platform with AI-driven Recommendations	React, Node.js, TensorFlow, MongoDB
Intelligent Blogging Platform with NLP-based Content Analysis	Python (Django, NLP), React
AI-Powered Online Learning Management System	Python (Django), React, GPT APIs
Crowdsourced Delivery Service with Real-time Tracking	React, Firebase, Google Maps API

Mobile App Projects

Project Title	Technologies Used
AI-Powered Budget Tracker	Flutter, Firebase, GPT APIs
Personalized Fitness Trainer using LLM	Java/Kotlin (Android), Swift (iOS), OpenAI API
Language Learning App with Real-time Speech Analysis	Flutter, Python (NLP, Speech-to-Text APIs)
AI Recipe Generator and Meal Planner	Swift, Firebase, GPT APIs
AI-Powered Mental Health Chatbot	React Native, Python (NLP, OpenAI API)

Game Development Projects

Project Title	Technologies Used
AI-driven NPC Behavior in 3D Open World Game	C# (Unity), TensorFlow
Agentic AI in Multiplayer Games	C++ (Unreal Engine), Reinforcement Learning
AI-Powered Dynamic Puzzle Generator	C# (Unity), Python (ML)
Sentiment-Responsive Interactive Storytelling Game	Unity, GPT APIs, NLP
Educational Quiz Game with Adaptive AI Tutor	Unity, Python (ML, Adaptive Learning)

Data Science & Machine Learning Projects

Project Title	Technologies Used
Stock Market Prediction using LLM	Python (Pandas, Scikit-Learn, GPT APIs)
AI-powered Resume Screening System	Python (NLP, OpenAI API)
Sentiment Analysis on Social Media Trends	Python (NLTK, TextBlob, BERT)
AI-driven Automated Legal Document Review	Python (NLP, LLM, GPT APIs)
Healthcare Predictive Analytics for Disease	Python (TensorFlow, Scikit-Learn, Medical

Risk Assessment	Datasets)
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IoT & Edge AI Projects

Project Title	Technologies Used
AI-powered Smart Home Automation System	Python, IoT Devices, Edge AI
AI-enabled Predictive Maintenance for IoT Devices	Python, TensorFlow, Raspberry Pi
AI-based Traffic Management System	Python, IoT Sensors, Reinforcement Learning
Smart Agricultural System with Real-time Monitoring	Python, IoT, AI-driven Crop Analysis
AI-powered Weather Forecasting Station	Python, IoT Sensors, Edge AI

Agentic AI & LLM-based Projects

Project Title	Technologies Used
Autonomous AI Assistant for Research & Writing	Python (LangChain, OpenAI APIs)
AI-driven Personalized Learning Tutor	Python (GPT-4, Reinforcement Learning, Adaptive Learning)
AI-powered Code Review & Debugging System	Python (GPT APIs, Static Analysis Tools)
Real-time AI-Powered Sentiment-Based Chatbot	Python (NLP, OpenAI API, Sentiment Analysis)
Agentic AI for Automated Financial Portfolio Management	Python (Deep RL, GPT APIs, Market Data)

Cybersecurity & Ethical AI Projects

Project Title	Technologies Used
AI-powered Phishing Detection System	Python (NLP, Cyber Threat Intelligence APIs)
AI-driven Fraud Detection System	Python (ML, Anomaly Detection)
Bias Detection and Mitigation in LLMs	Python (NLP, Fairness & Ethics APIs)
AI-Powered Digital Forensics for Cybercrime Investigation	Python, Cyber Threat Intelligence
AI-based Secure Authentication System	Python, Biometric Data, LLM